Immune regulation

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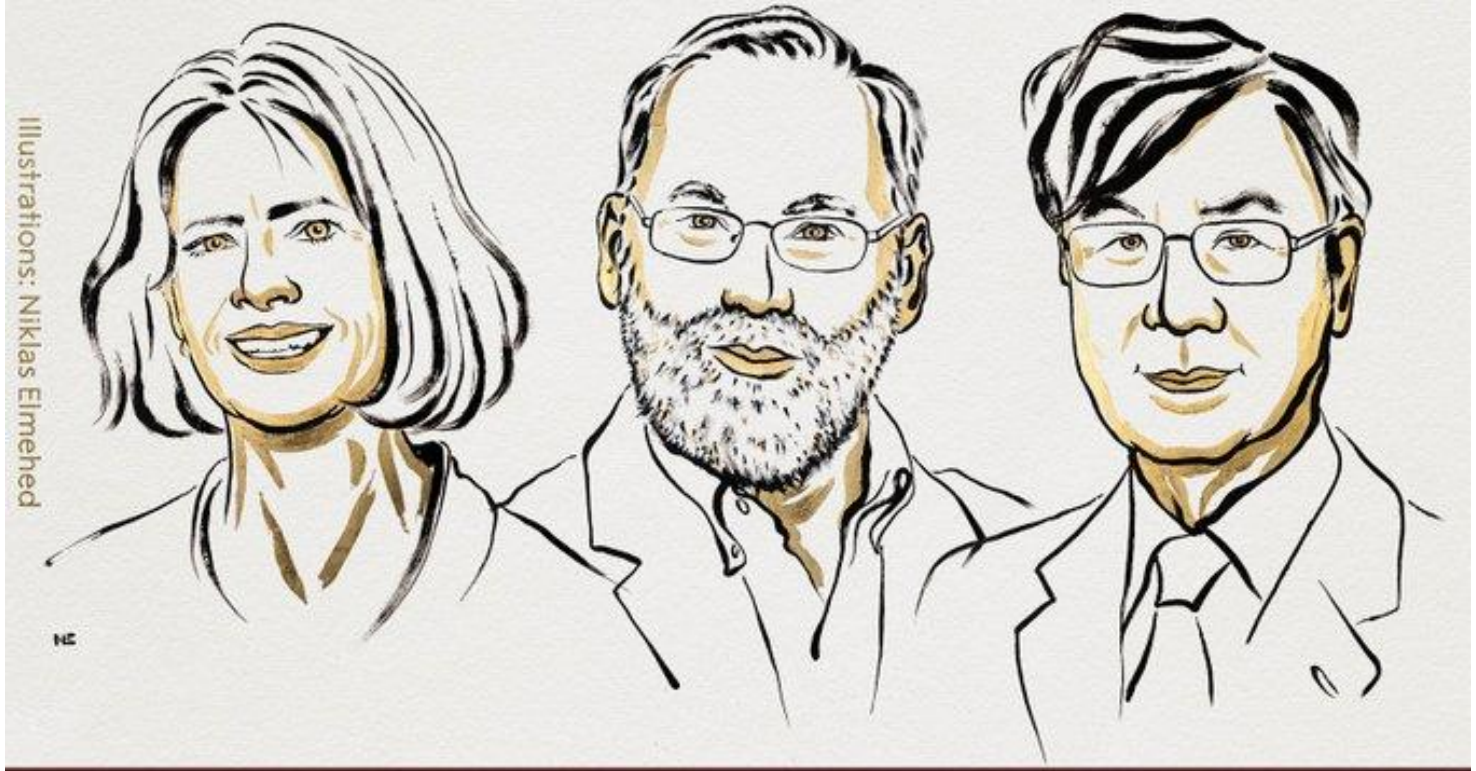
The Immune System

- Efficient immunity against foreign antigens to protect the host from harmful pathogens with minimum pathology
- Distinguish innocuous foreign antigens
- Be tolerant to self antigens to prevent autoimmunity

Tolerance and regulation

- Central Tolerance
  - in the thymus: T cell deletion of potentially autoreactive cells (negative selection)
  - In the bone marrow: B cells (clonal deletion, receptor editing)
- Peripheral tolerance
  - lymph node, lymphoid organ, or other tissues
  - self-Ag, innocuous Ag

THE NOBEL PRIZE IN PHYSIOLOGY OR MEDICINE 2025



Illustrations: Niklas Elmehed

Mary E. Brunkow

Fred Ramsdell

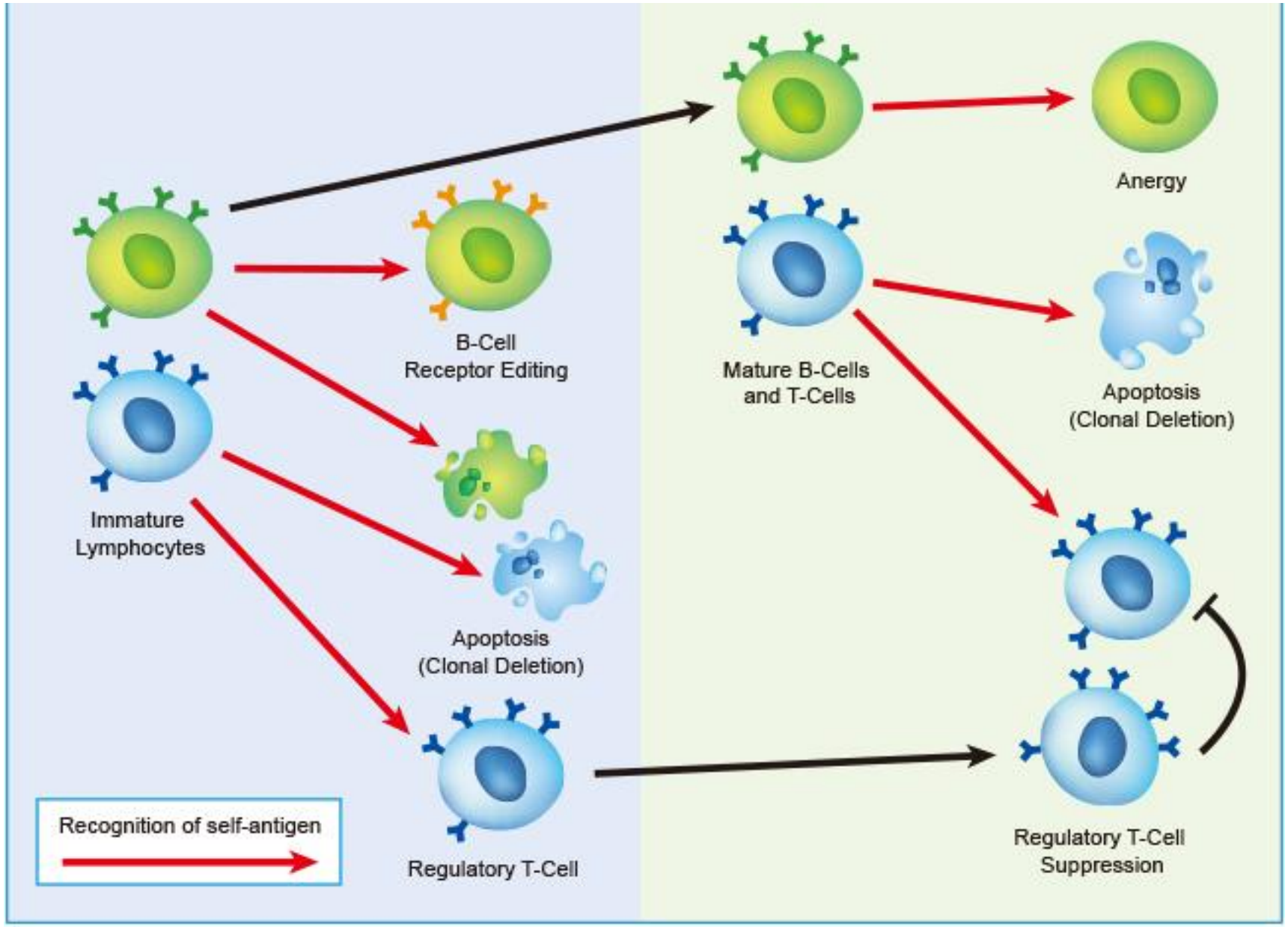
Shimon Sakaguchi

"for their discoveries concerning peripheral immune tolerance"

THE NOBEL ASSEMBLY AT KAROLINSKA INSTITUTET

Central Tolerance:
Thymus, Bone marrow

Peripheral Tolerance:
Lymph node, peripheral tissue



Peripheral tolerance

- Central tolerance is not perfect
- Self-reactive T cells present in periphery

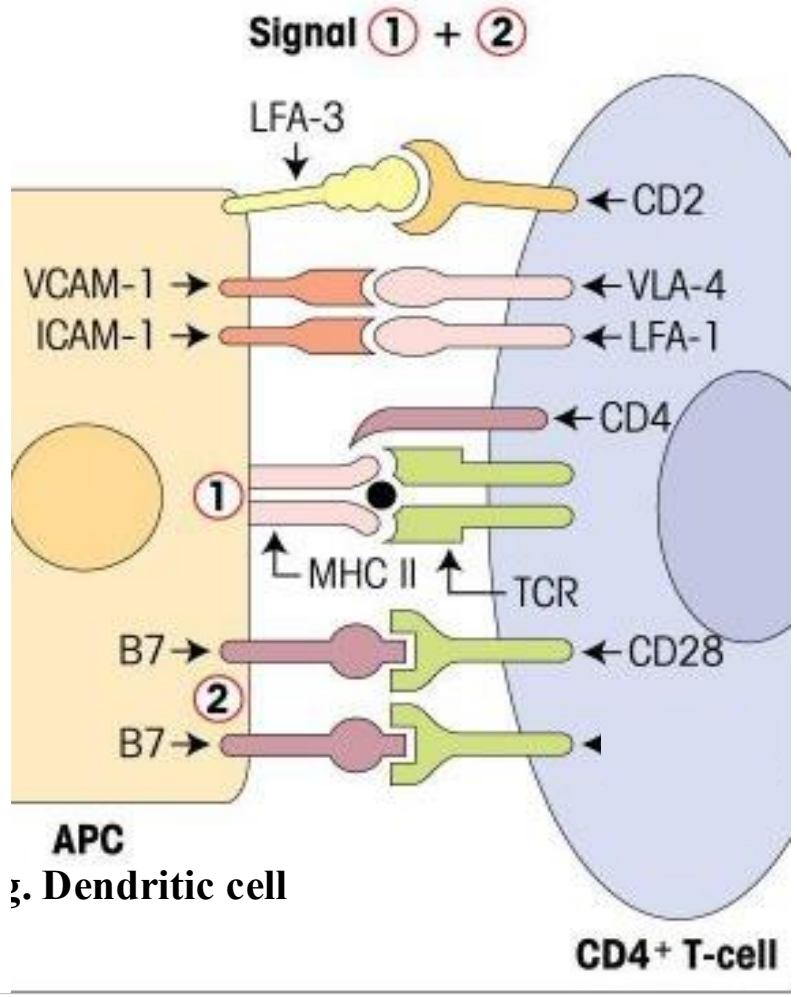
- Peripheral tolerance
  - Prevent autoimmunity by preventing activation of self reactive cells
  - Prevent responses to microbiota/ food/ innocuous antigens
  - Regulate magnitude and timing of protective immune responses

Mechanisms of peripheral tolerance and immune regulation

- Anergy
- Deletion/ Apoptosis
- Immune checkpoints
- Active regulation (Tregs, Bregs, regulatory cytokines)

- Ignorance/ quiescence

T cell activation

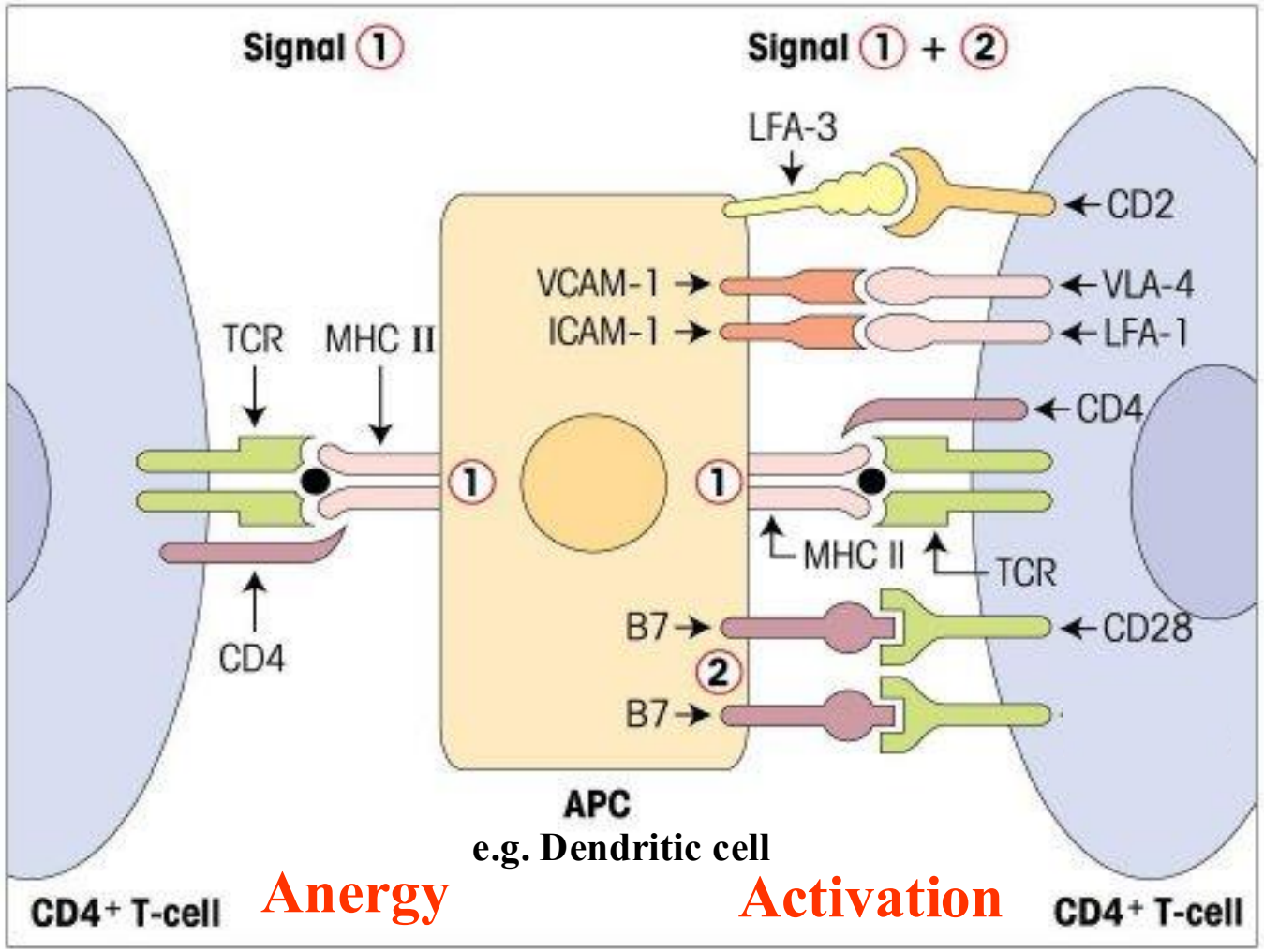


g. Dendritic cell

Activation

Delves *et al. Roitt's Essential Immunology, 12th ed.*
© 2011 Delves *et al.* Published 2011 by Blackwell Publishing Ltd.

T cell anergy



Delves *et al. Roitt's Essential Immunology, 12th ed.*
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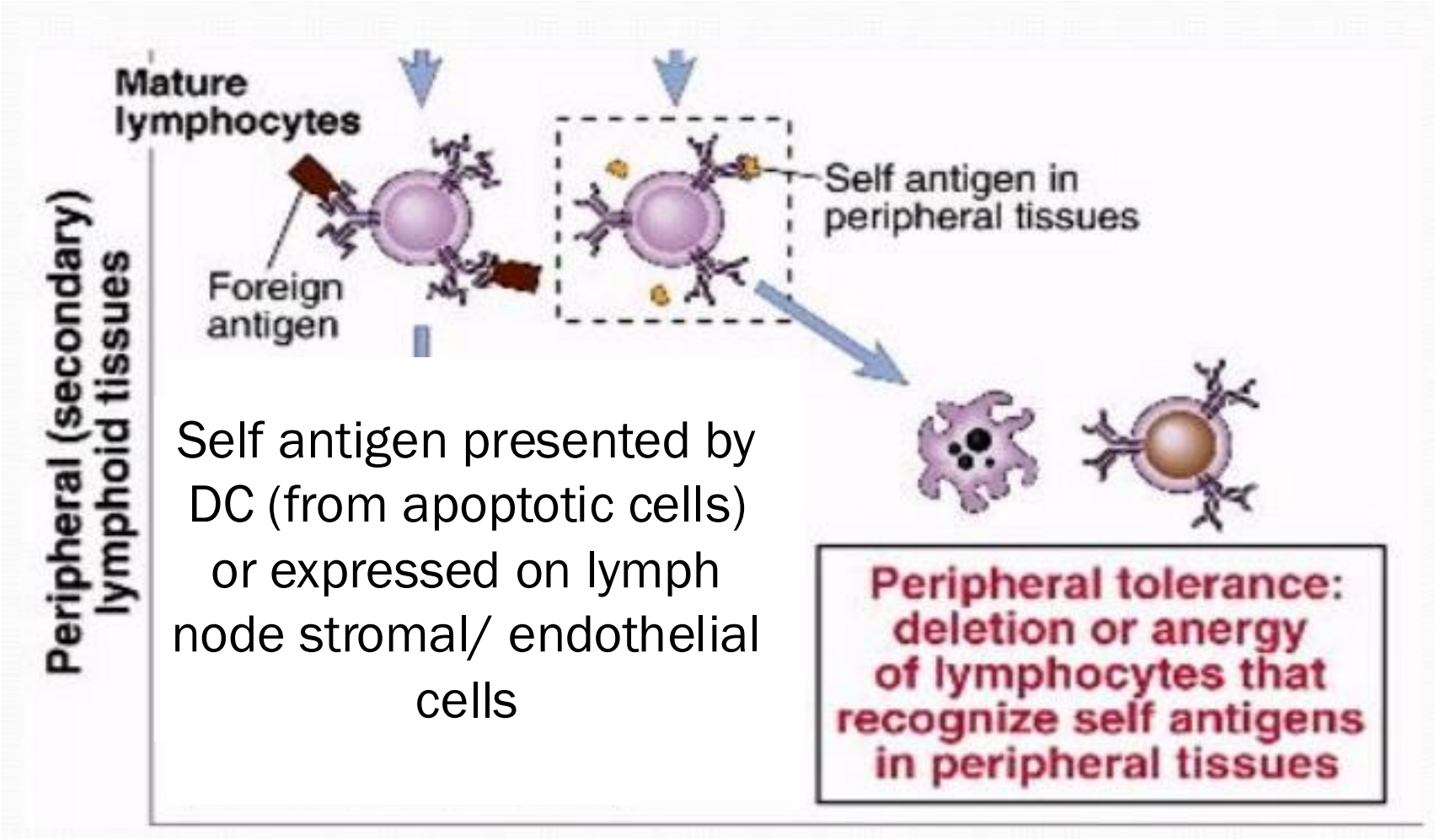
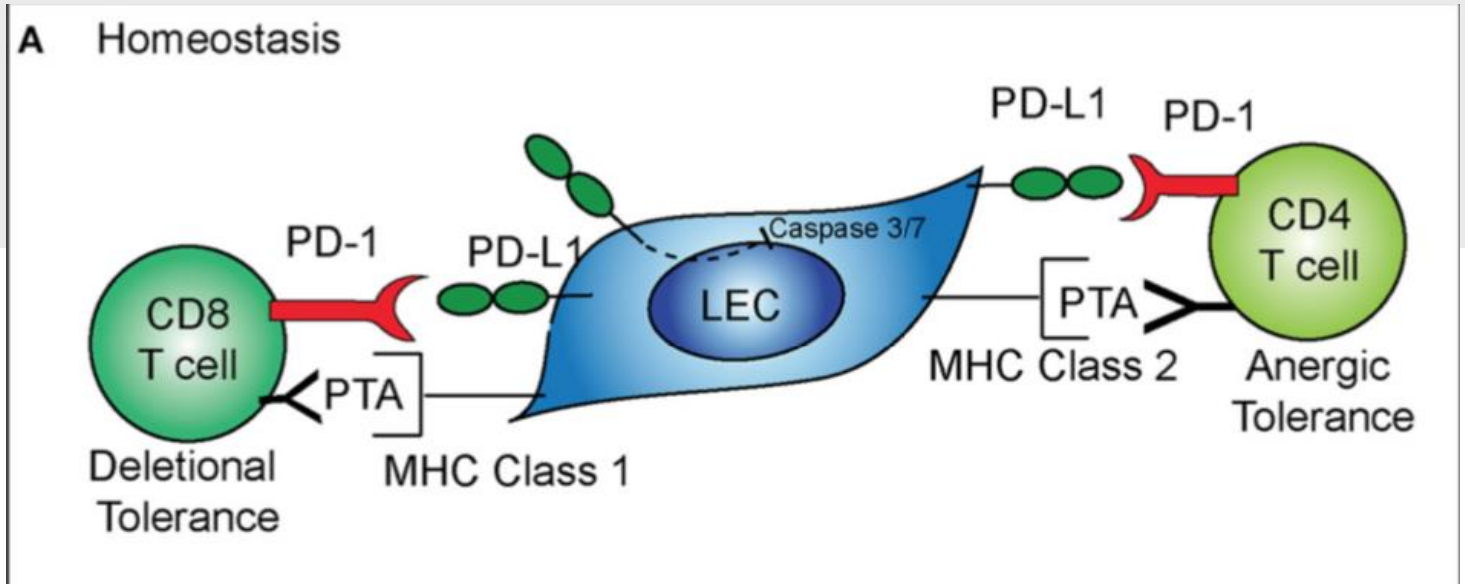
Peripherally induced tolerance



Self antigen presented by DC (from apoptotic cells) or expressed on lymph node stromal/ endothelial cells

Lymph node stromal cells in peripheral tolerance

- Expression of peripheral tissue antigen (Aire dependent/ independent)
- Fibroblastic reticular cells (FRC) deletional tolerance in CD8/ anergy CD4
- Lymphatic endothelial cells (LEC): lack co-stim molecules; can induce PDL1:PD1 pathway



Lucas and Tamburini

Front Immunol 2019

Apoptosis in Immune Regulation

Programmed cell death via apoptosis: limits the pool of effector T cells

- Activation induced cell death (AICD): a built-in negative feedback mechanism to prevents excessive lymphoproliferation and immunopathology and maintain homeostasis: FAS-FASL interaction

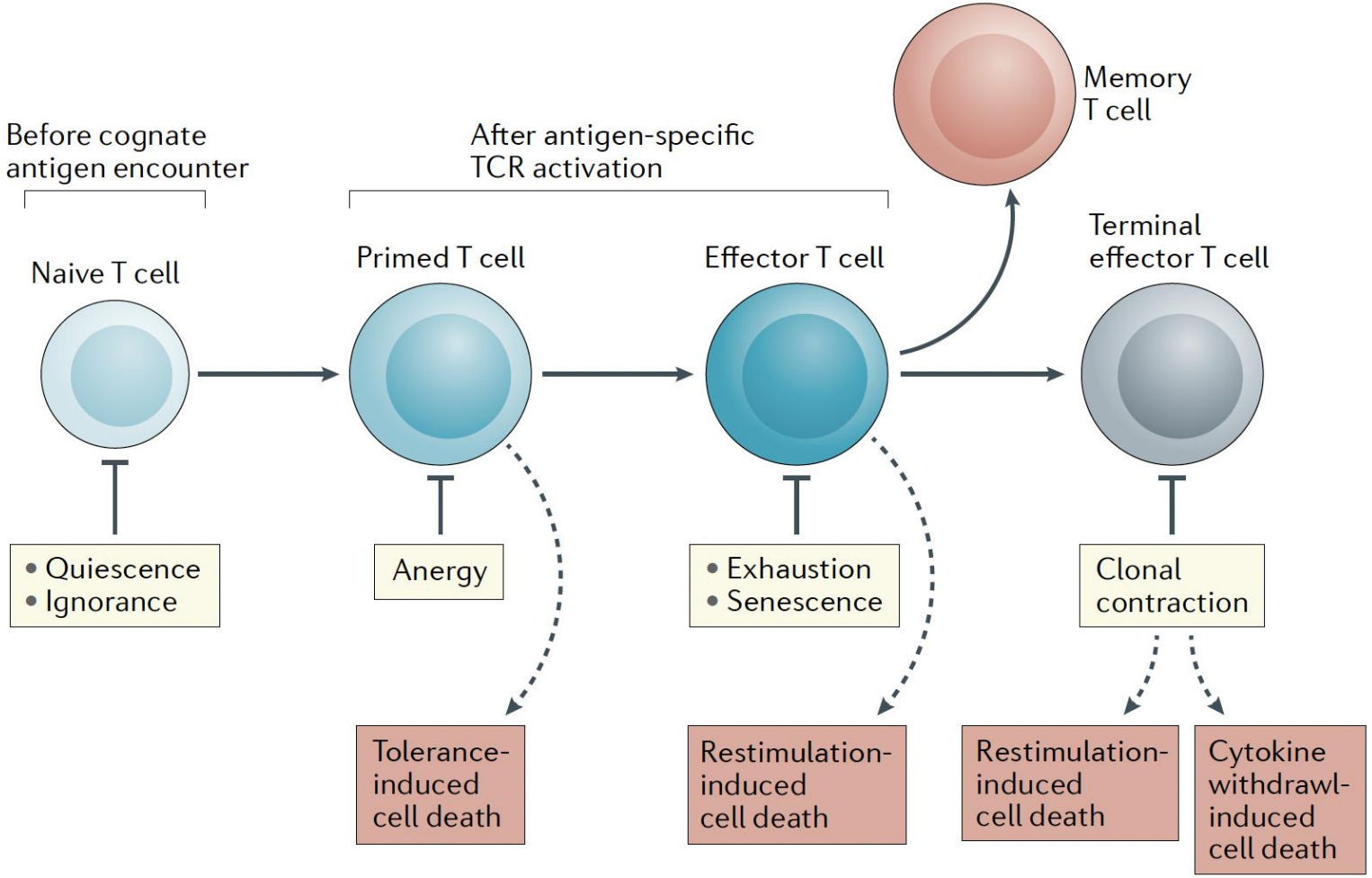
- intrinsic apoptosis pathway triggered by cytokine withdrawal: major contraction phase of the T cell response; only a small fraction of T cells survive and later constitute immunological memory.

Fas- FasL in peripheral tolerance

- Activated T cells express Fas and FasL
- Killing through suicide or fratricide
- Deals with over-activated effector cells
- Important in preventing autoimmunity by removing self-reactive cells
  - Autoreactive T cells ↑
  - Fas KO (lpr/lpr) and FasL KO (gld/gld) mice
  - ALPS patients

Yamada et al. Front Immunol 2017:8: 403

T cell tolerance mechanisms: integrated view

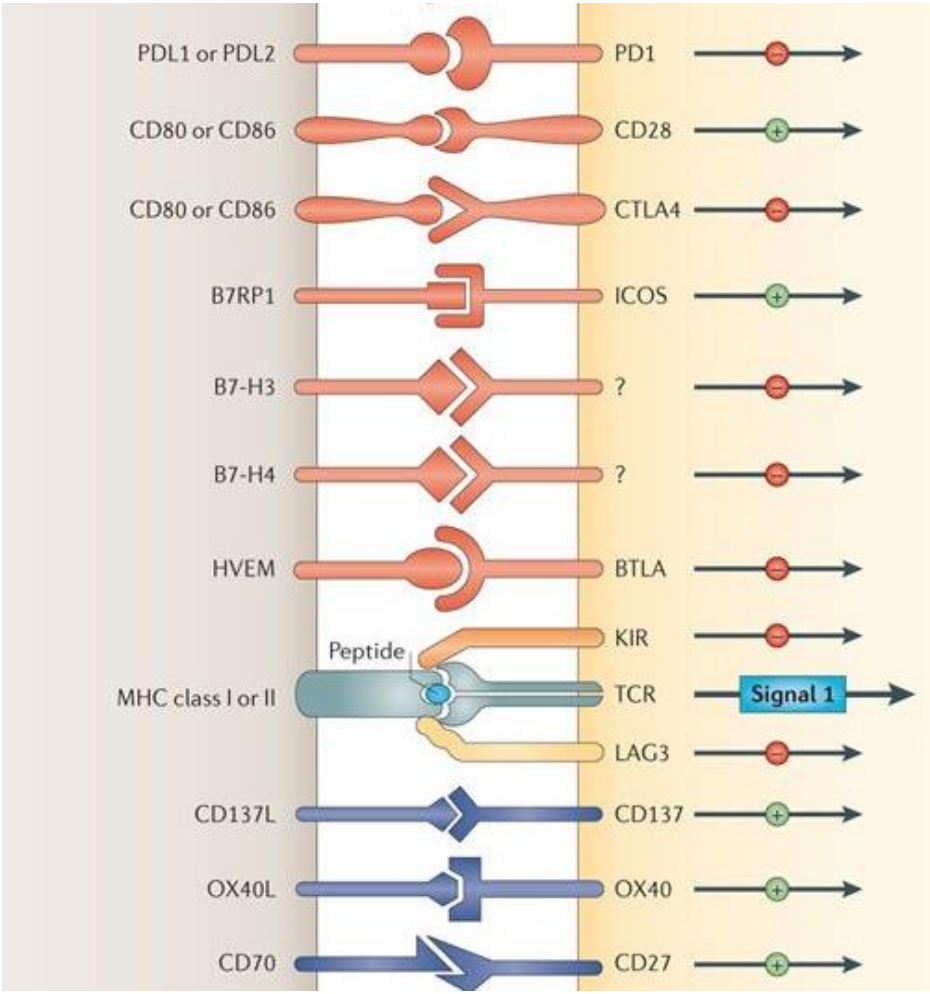


Eltanbouly & Noelle Nat Rev Immunol 2020

Regulating immune responses: immune checkpoints/ inhibitory receptors

- Expression of co-inhibitory molecules on effector T Cells (PD1, CTLA4): intrinsic control of T cells and is critical for the maintenance of peripheral tolerance/homeostasis
- Immune-inhibitory enzymes (indoleamine 2,3-dioxygenase (IDO), arginase)
- Functional exhaustion and loss of effector functions (especially in chronic stimulation)

Stimulatory and Inhibitory receptors



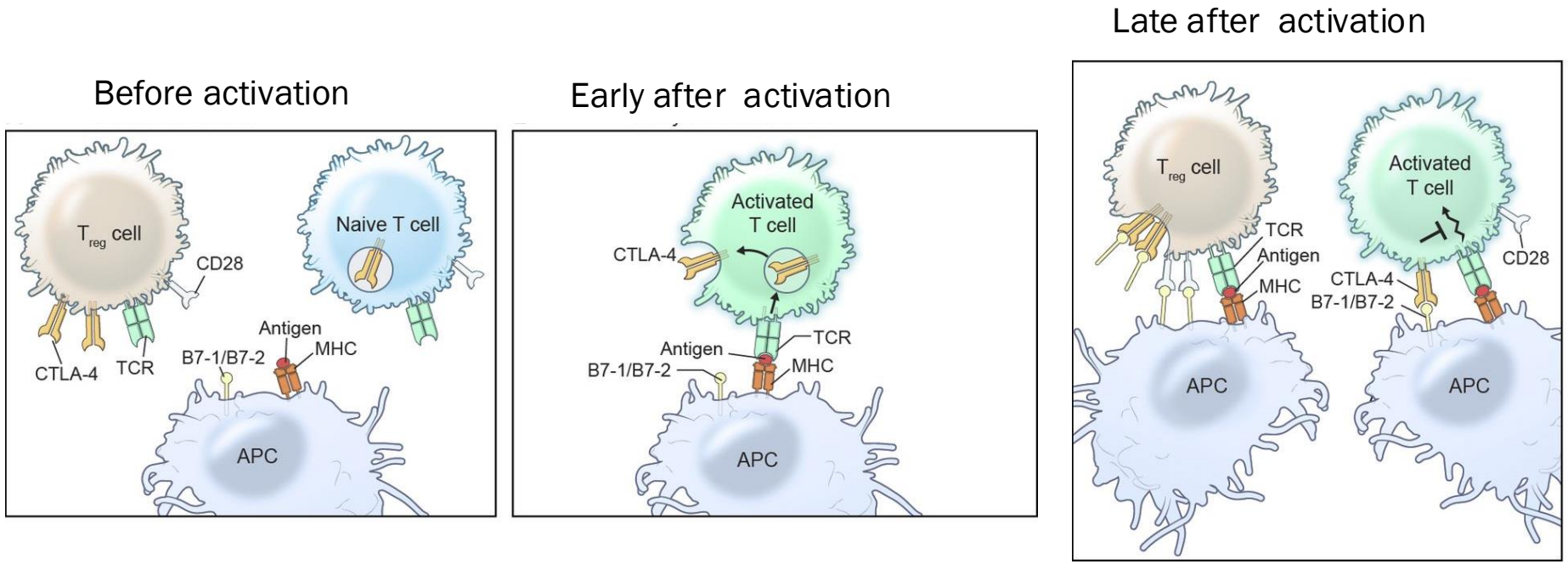
CTLA4

- CTLA-4 is an inhibitory relative of the T cell co-stimulatory molecule CD28
- Interact with the same shared ligands CD80 (B7.1) and CD86 (B7.2) (20x higher affinity)
- critical inhibitor of autoimmunity : CTLA4 KO lethal (huge lympho-proliferation/ over activation of T cells)
- CTLA4 polymorphisms associated with autoimmunity

CTLA-4: Mechanism of action

- Limits TCR signaling (counteracts TCR/CD28 signaling); mainly in 2nd lymphoid organs (activation of naive T cells)
- Competition with CD28; context sensitive: relative amount of ligands available ('danger' upregulates CD80/ CD86 on APCs)
- CTLA-4 can physically remove its ligands from antigen presenting cells by a process of trans-endocytosis
- Signaling through CTLA-4: possibly activation of SHP2, dephosphorylates CD3zeta chain (disputed as the key mechanism)

Mechanisms of CTLA-4 function

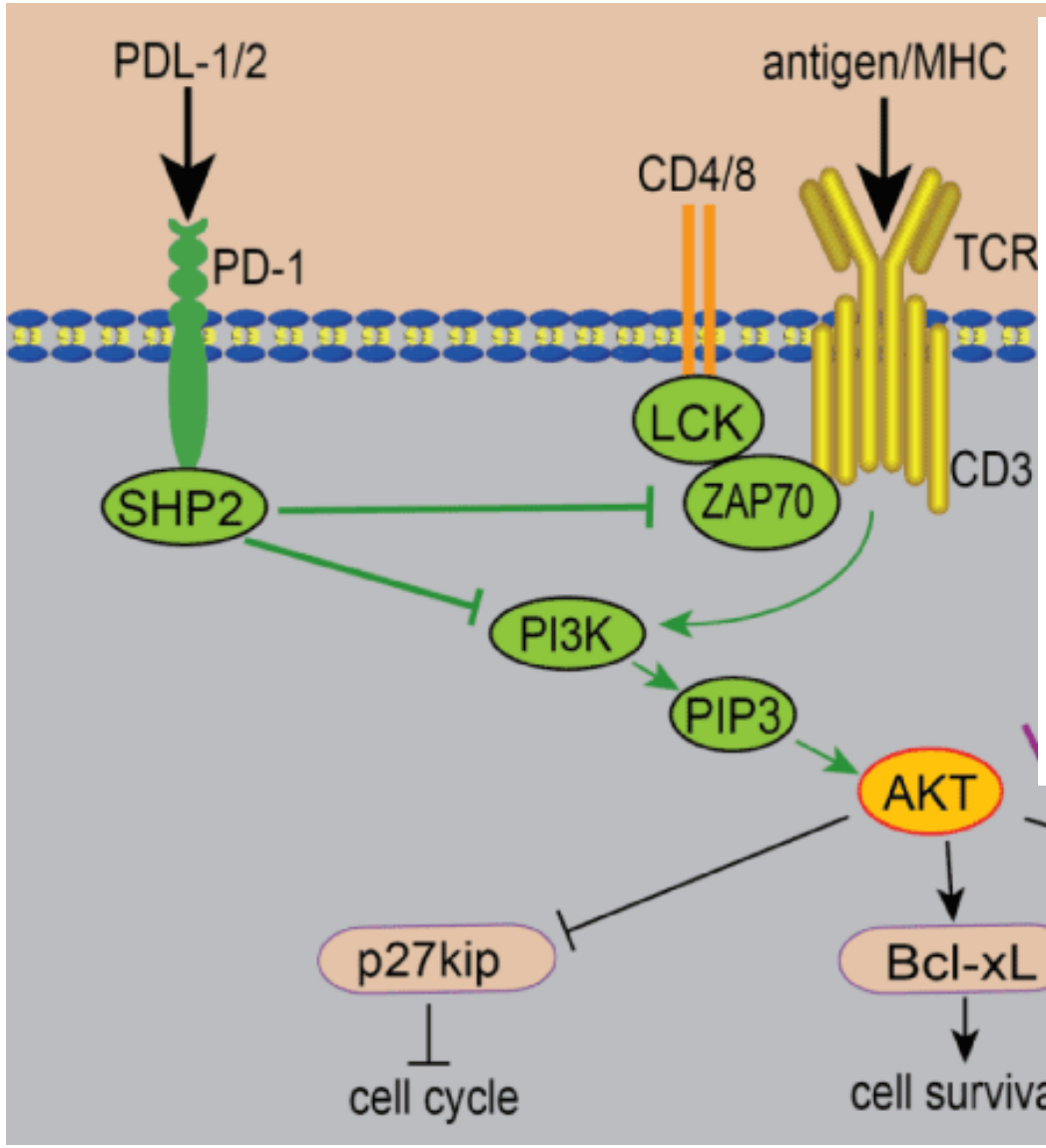


- CTLA-4 is up-regulated on activated T cells; inhibits further T cell activation by out-competing CD28
- Treg cells, which constitutively express CTLA-4, trans-endocytose B7 ligands on APCs to prevent CD28 co-stimulation

Jill M. Fritz and Michael J. Lenardo J Exp Med 2019;216:1244-1254

PD1

(Programed cell death protein 1)



- Predominantly controls effector T cells in tissues
- Often associated with chronic activation/ exhausted cells
- Can also induce anergy

Other inhibitory co-receptors

- Tim3, Lag3, TIGIT
- Up-regulated on activated cells, especially on chronically activated cells (exhausted/ dysfunctional)
- Implicated in autoimmune diseases and in tumor responses/ chronic viral infections
- Negative regulation of T cell expansion

Eltanbouly & Noelle Nat Rev Immunol 2020 https://doi.org/10.1038/s41577-020-00454-2
Joller, Anderson and Kuchroo Immunity 2024 https://doi.org/10.1016/j.immuni.2024.01.010

Immune regulation: immune checkpoints & therapeutic approaches

- Checkpoint inhibition to mobilise immune responses in cancer
  - Antibodies to CTLA-4, PD-1, PD-L1
  - Combination therapies (Tim3, Lag3, TIGIT)
- Targeting checkpoint in autoimmunity: activate inhibitory receptors?
  - CTLA4-Fc (RA, Juvenile arthritis)

Conclusions part I

- Self reactive T cells:
  - Ignorance/Quiescence
  - Anergy
  - Deletion/ apoptosis

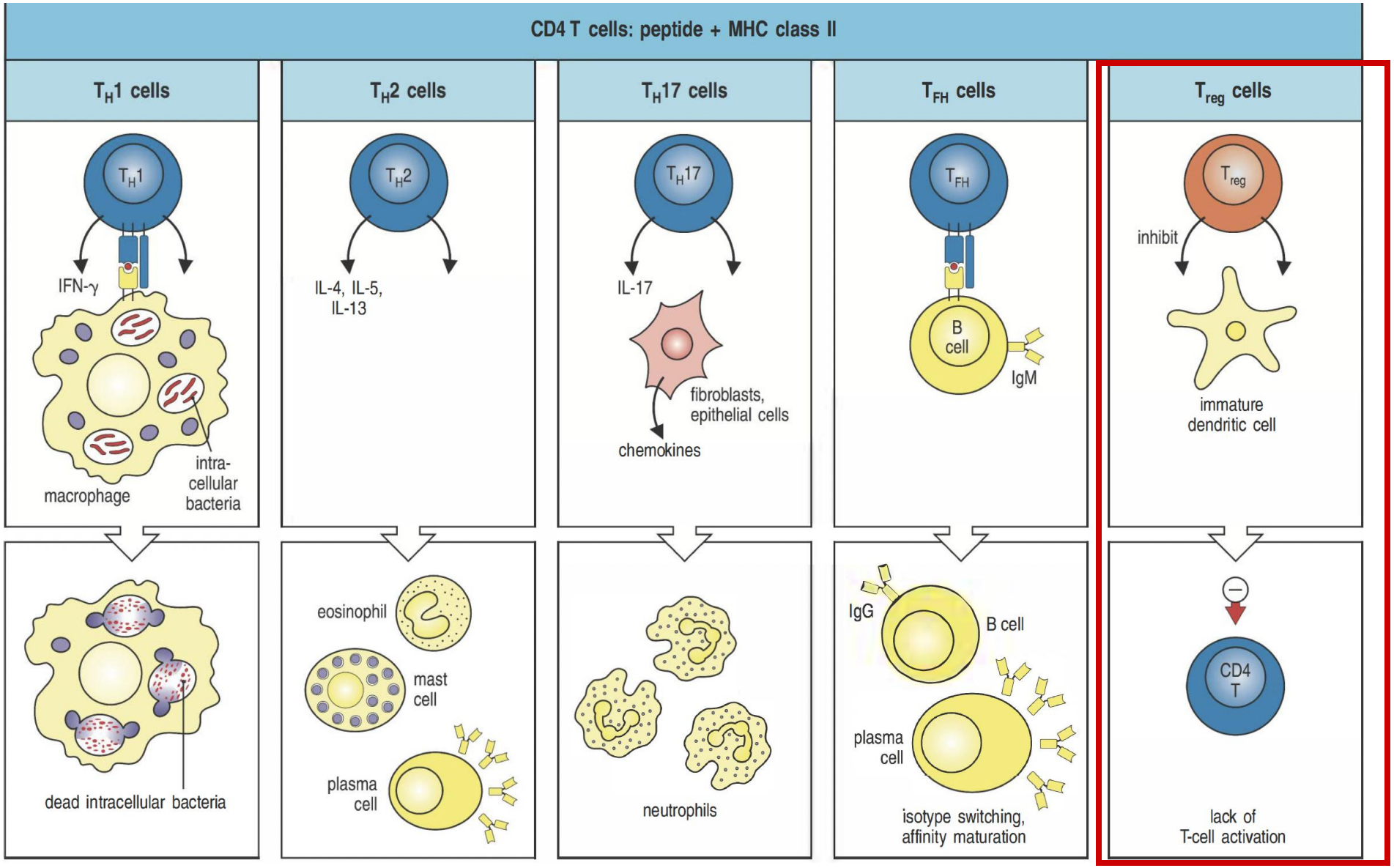
- Regulation of T cell responses:
  - Deletion/ apoptosis
  - Checkpoint molecules
  - Exhaustion

Eltanbouly & Noelle Nat Rev Immunol 2020 https://doi.org/10.1038/s41577-020-00454-2

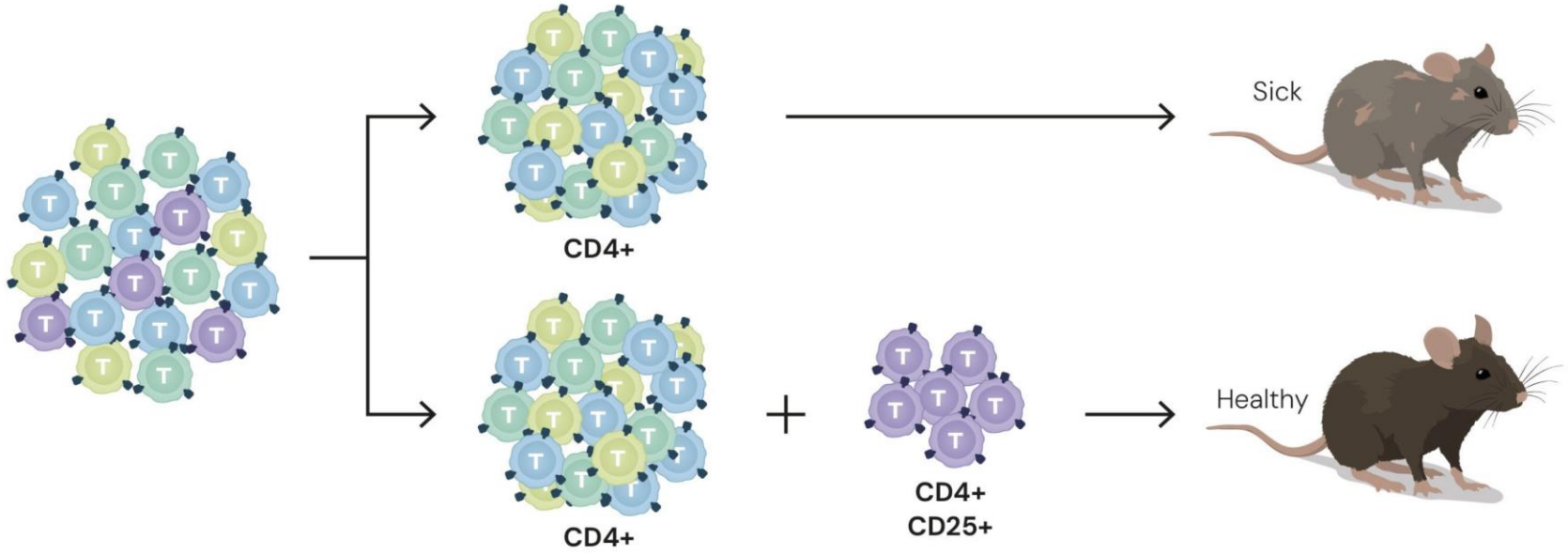
Immune regulation part 2:

Regulatory T cells

Regulatory T cells (Treg)



Subset of T cells specialised in suppression of immune responses



Sakaguchi defines a new class of T cells

Sakaguchi showed that the T cells with CD25 on their surface protect against autoimmune diseases through an experiment in mice that lacked T cells. If he injected CD4-bearing T cells into the mice, but removed all the cells with CD25, the mice developed serious autoimmune diseases. If he added CD25-bearing cells, the mice remained healthy.

1995

Regulatory T cells (Treg)

- $CD4^{+}CD25^{+}Foxp3^{+}$
- 5-10% of peripheral CD4 T cells in mice and humans
- Lymphoid organs as well as peripheral tissues
- Expression of CTLA4
- Lack of IL-2 production

Characteristics of Foxp3^{+} Tregs

Transcription factor	Activation and memory	Homing and origin	Suppressive and effector function	Apoptosis, survival or other
FOXP3	CD45RA CD45RO CD25 HLA-DR Lack of CD127 CD69	CD62L CCR4 CCR6 CCR9 CD103 CD304 CD31 Lack of CD49d	CTLA4 ICOS CD39–CD73 LAP Granzyme B Galectin 1 Galectin 10 TRANCE CD80 and CD86 IL-10 IL-17 CD2 Lack of IL-2	CD27 OX40 CD95 PD1 GITR Galectin 3 GARP MS4A4B IL-1R CD6

Role for Regulatory T cells

- prevent immune responses to self Ag
- prevent responses to innocuous Ag/ microbiota
- control the magnitude of protective immune responses (bacteria, viruses) to prevent tissue damage

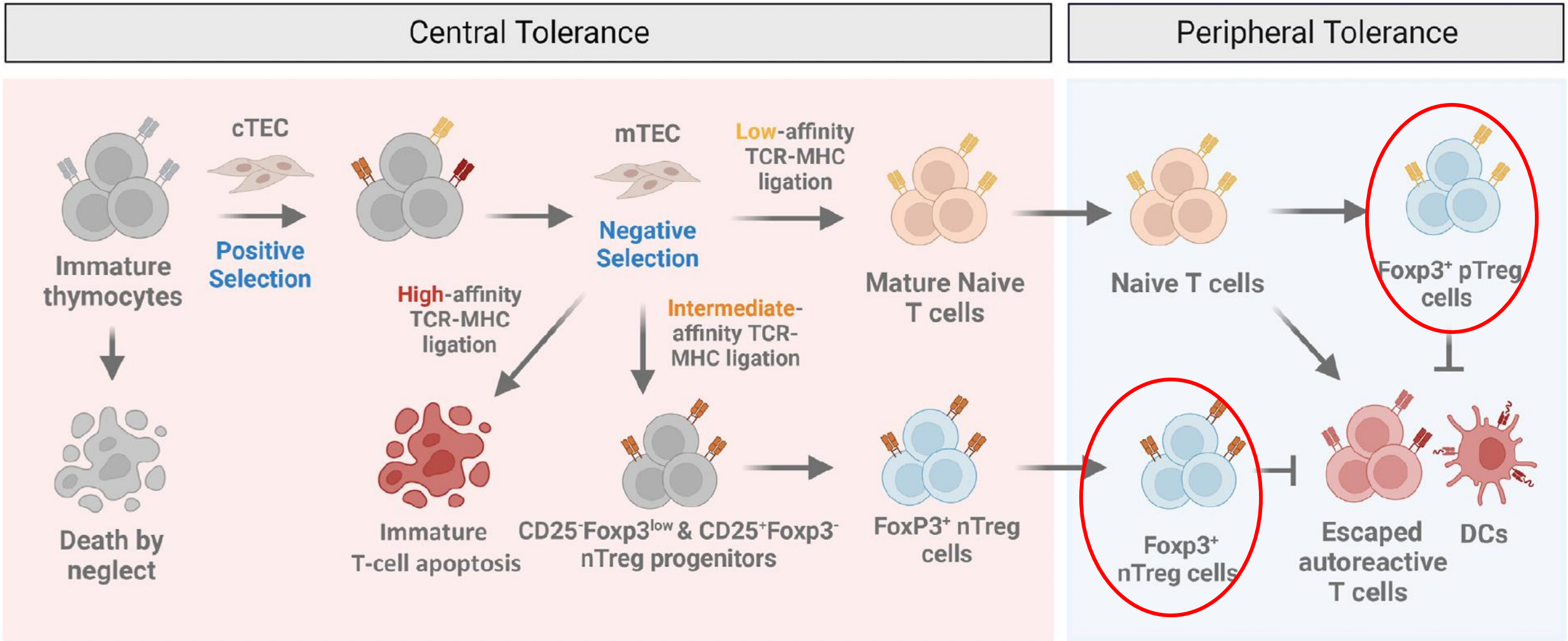
- **Involved in tissue homeostasis**
  - Adipose tissue resident Tregs involved in metabolic regulation
  - Muscle resident Treg involved in muscle repair
  - Colonic Tregs

for review see: 'Tissue Tregs' M Pandurro, C Benoist & D Mathis Annu. Rev. Immunol. 2016. 34:609–33

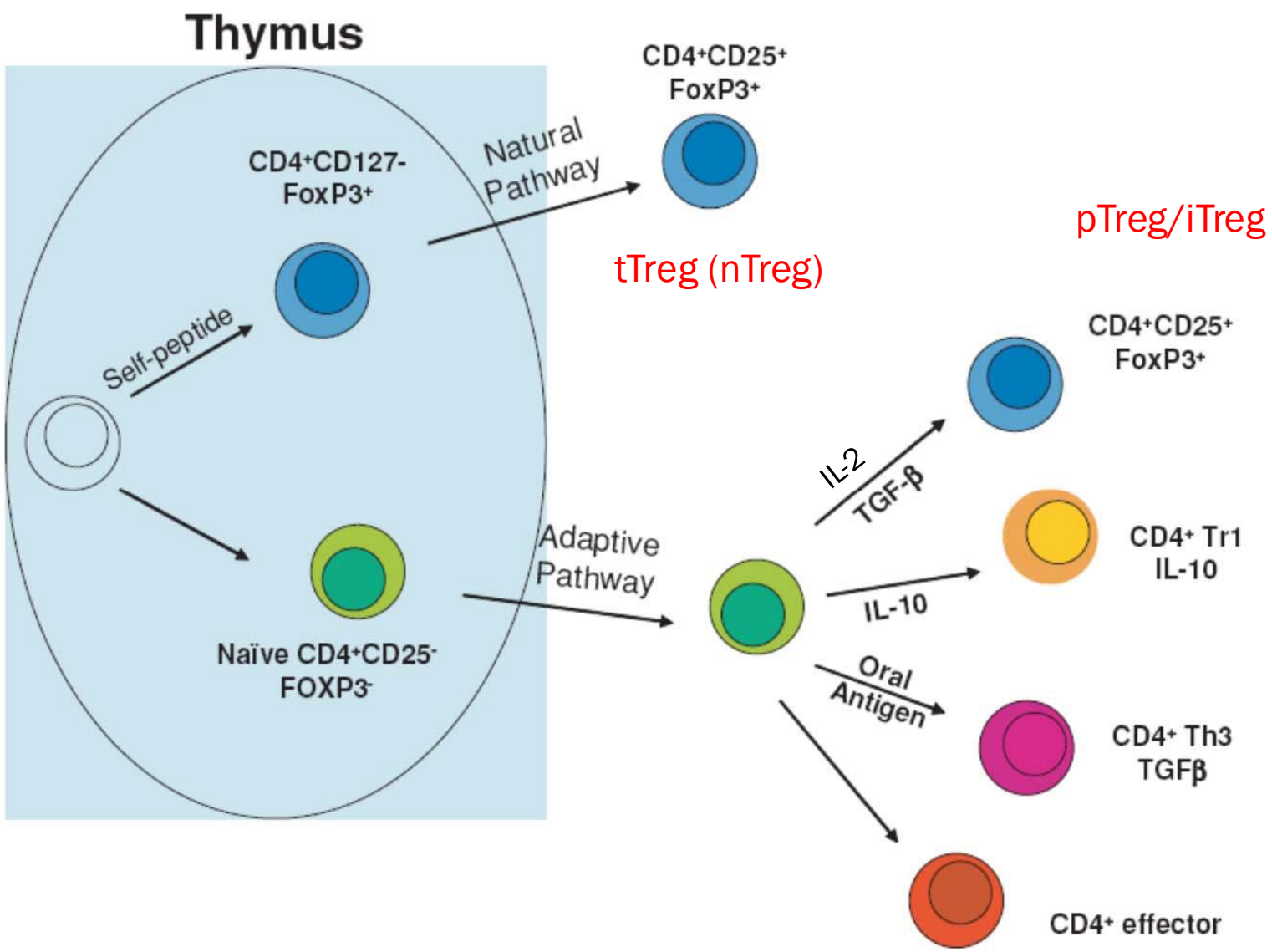
Regulatory T cells: Questions

- How many types exist and how can they be distinguished from other T cell populations?
- Where and how are Tregs generated?
- What is their mechanism of action and targets?
- Can they be manipulated for clinical benefit?

T cell tolerance



Types of Regulatory T cells



Types of $CD4^+CD25^+Foxp3^+$ Regulatory T cells

1. Thymus derived Tregs (**tTreg**)
   - Generated in the **thymus** as a separate lineage with regulatory function
   - Demethylation of "Treg cell-specific demethylated region" (TSDR) in the Foxp3 enhancer region
   - Expression of Helios and Neuropilin
2. Peripherally induced Tregs (**pTreg/ iTreg**)
   - Tregs induced in the periphery from conventional CD4 T cells
   - Less stable: TSDR may not be fully demethylated

2001

- specific mutation to a gene they named Foxp3 makes the mouse strain they were studying particularly vulnerable to fatal autoimmune diseases (Scurfy)

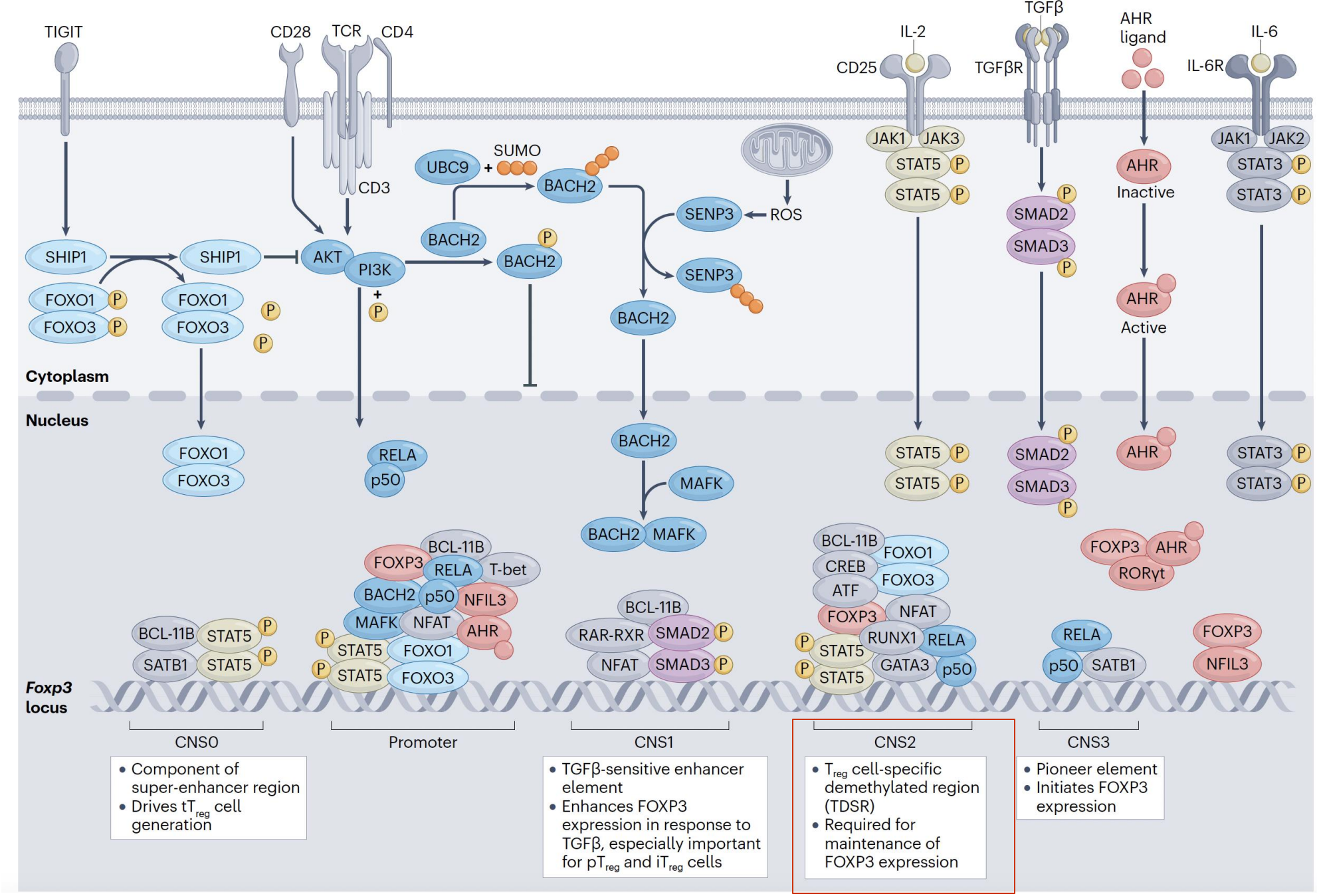
- similar mutations caused a devastating autoimmune disease known as IPEX syndrome in humans.

Foxp3 and Tregs: master regulator?

- Forkhead/winged-helix transcription factor (X-chromosome encoded) which is used for a molecular definition of $CD4^+CD25^+$ Tregs
- Deletion/ inactivation of Foxp3 gene; hyper-activation of T cells (scurfy mice), autoimmune disease (IPEX patients)
- Foxp3 controls the development of Tregs and their function: induction or forced expression of the Foxp3 gene in normal naive T cells is able to convert them to 'Treg-like' cells with *in vivo* and *in vitro* suppressive function
- Foxp3 is likely to control the expression of key molecules mediating suppression (CTLA-4, CD25)

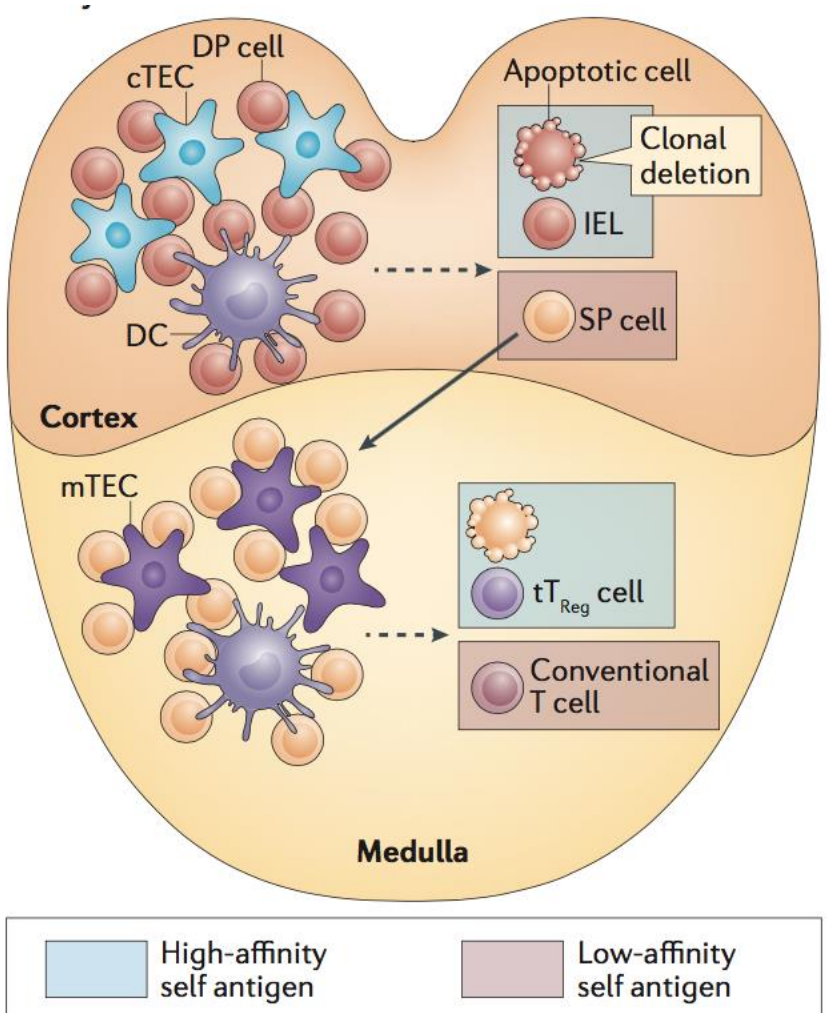
Foxp3 expression≠ regulatory function?

- Human T cells can express Foxp3 transiently after activation, this does not enable suppression
- Human Foxp3$^+$ cells can lose Foxp3 expression and the ability to suppress in culture
- Conventional mouse CD4$^+$ cells readily convert into Foxp3$^+$ ‘induced’ Tregs with similar surface markers but different gene expression profile (lack Treg signature, TSDR region)
- Human Foxp3$^+$ pTregs little suppressive ability but can express inflammatory cytokines (plasticity?)



Trujillo-Ochoa et al. The role of transcription factors in shaping regulatory T cell identity. *Nat Rev Immunol* (2023).

Generation of tTreg



- $CD4^+CD25^+Foxp3^+$ cells with regulatory function exist in the thymus (5% SP thymocytes express CD25)
- Thymectomy (few days after birth) leads to autoimmunity
- Recognition of antigen with intermediate affinity (too low for deletion, too high to allow migration to the periphery as conventional T cells: become Treg)

Generation of Foxp3$^+$ Regulatory T cells in the periphery

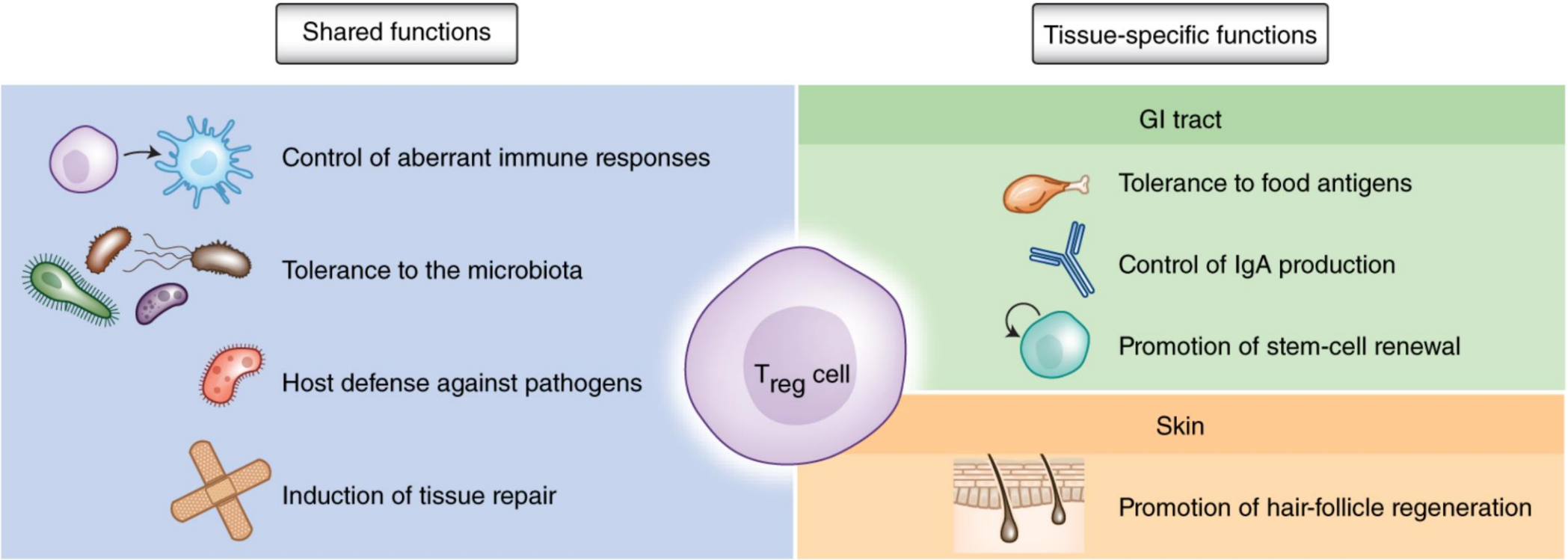
- In vitro generation of CD4$^+$CD25$^+$Foxp3$^+$ from CD4$^+$CD25$^-$ responder cells eg. in the presence of TGF-β
- In the presence of 'tolerogenic' conditions: suboptimal co-stimulation, ⇧TGFβ, ⇧Retinoic acid
- Selective depletion of pTreg leads to late onset allergic like inflammation in the gut/ lung
- Specificity: commensals, dietary antigen, foreign antigen (in contrast tTreg: self-antigens)
- May not be as stable as tTreg: TSDR region not fully demethylated?

Treg specialisation

- Different subsets specialised for suppression of specific Th subsets: some shared genes/ chemokine receptors; More effective?
- Visceral Adipose tissue: Tregs prevent abnormal inflammation (obesity, T2D, Atherosclerosis); express PPARγ
- Gut: microbiota (mainly pTreg), pathogens, tissue repair
- Skin: cutaneous wound healing, express CCR4, CCR6, hair follicles, microbiota
- Muscle/Lung: injury response, tissue repair (amphiregulin)
- Increased in tumors

for review see: 'Regulatory T cell adaptation in the intestine and skin' Whibley et al
Nat Rev Immunol. 2019. 20:386-396

Tissue adaptation of Treg



for review see: 'Regulatory T cell adaptation in the intestine and skin' Whibley et al
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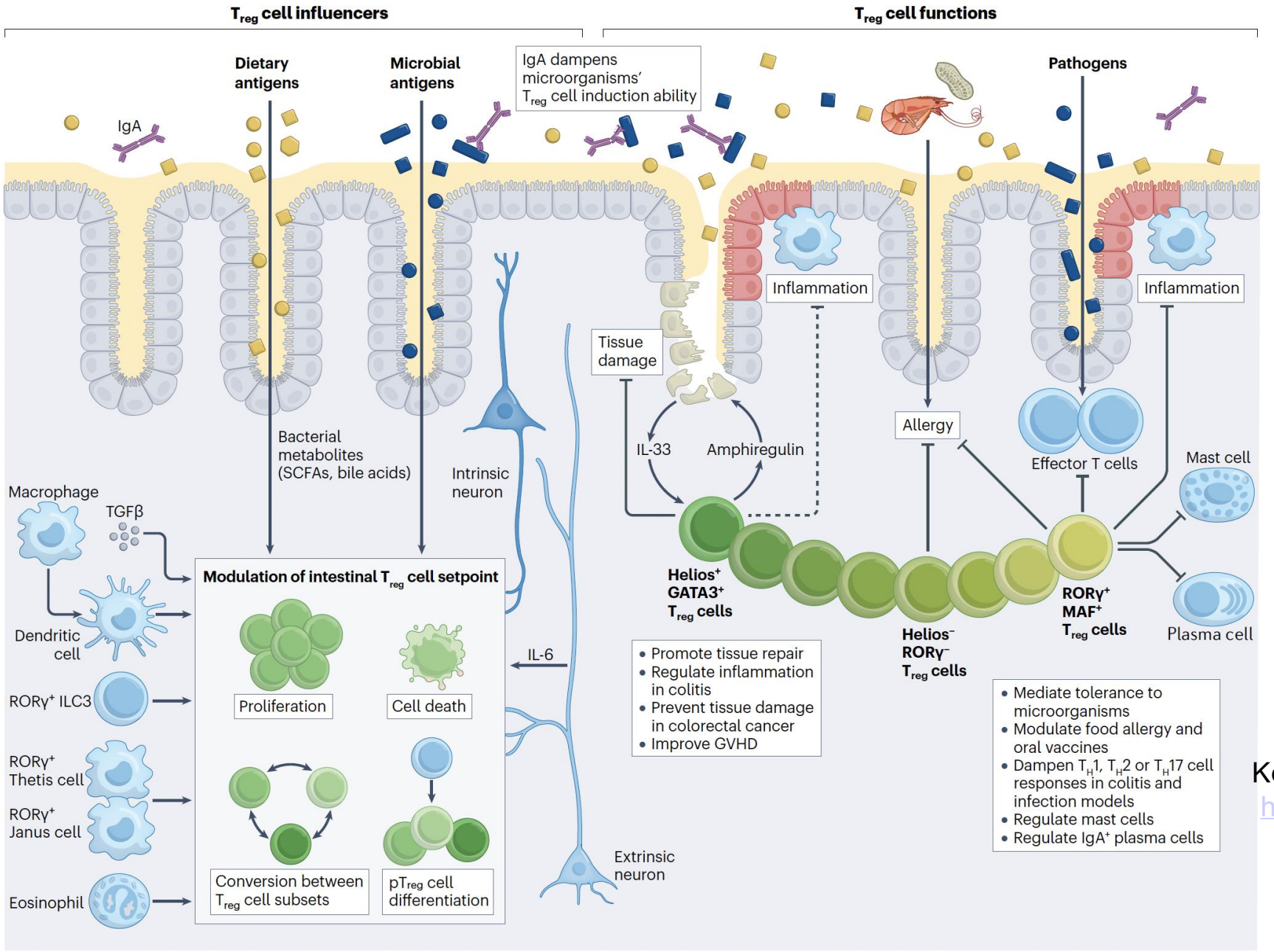
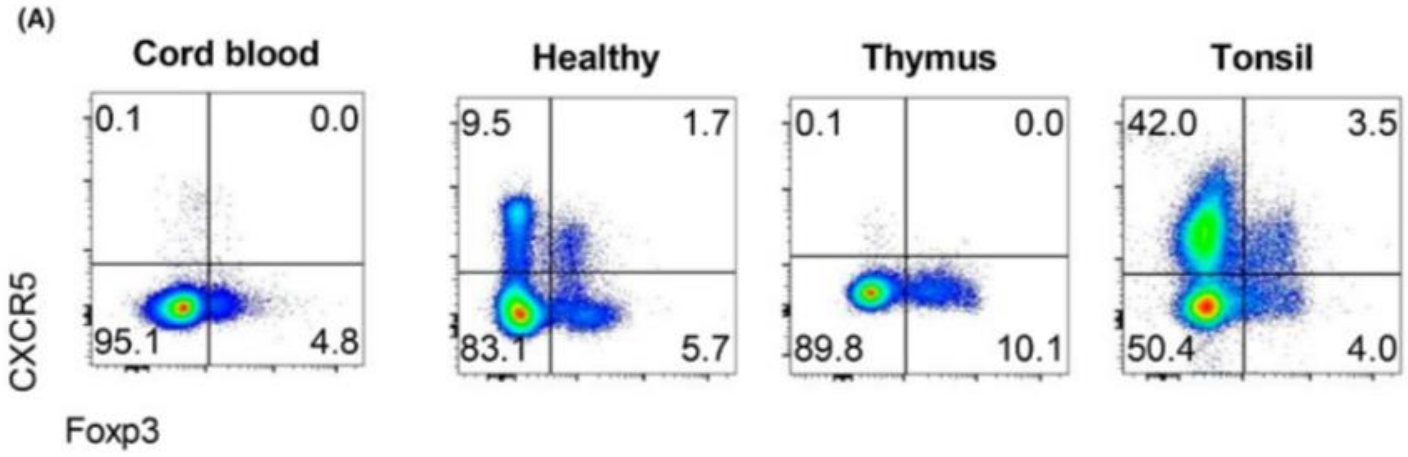
Intestinal regulatory T cells



Ramanan D et al 2023 Nat Rev Immunol Volume 23 | 749–762
https://doi.org/10.1038/s41577-023-00890-w

Kedmi & Litman (2024) Immunity, 57,10, 2269
https://doi.org/10.1016/j.immuni.2024.09.011

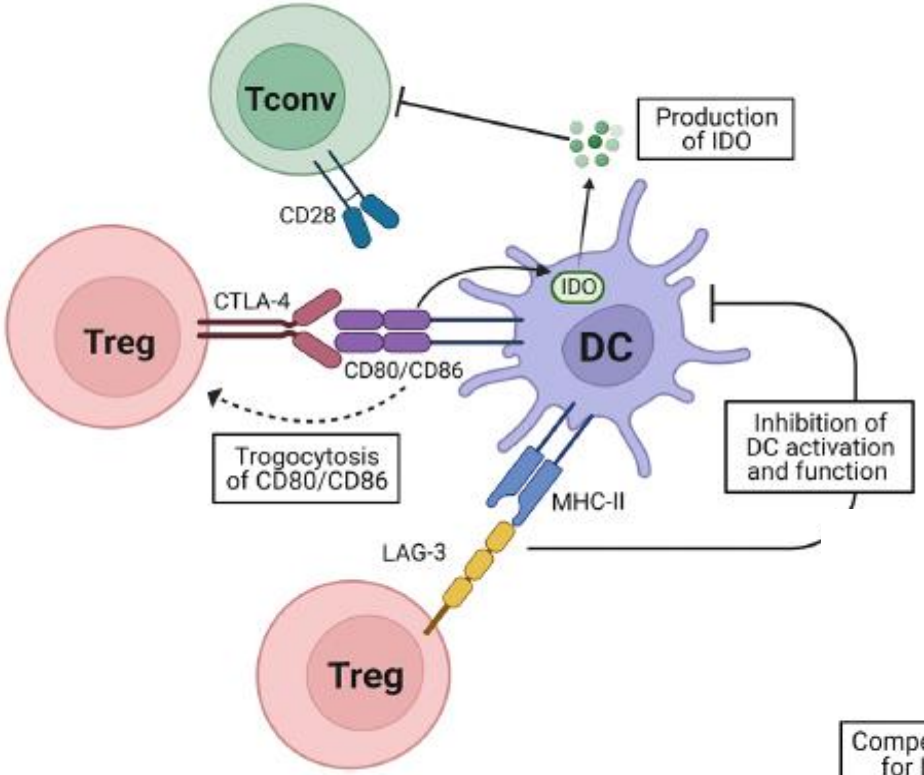
T follicular regulatory cells

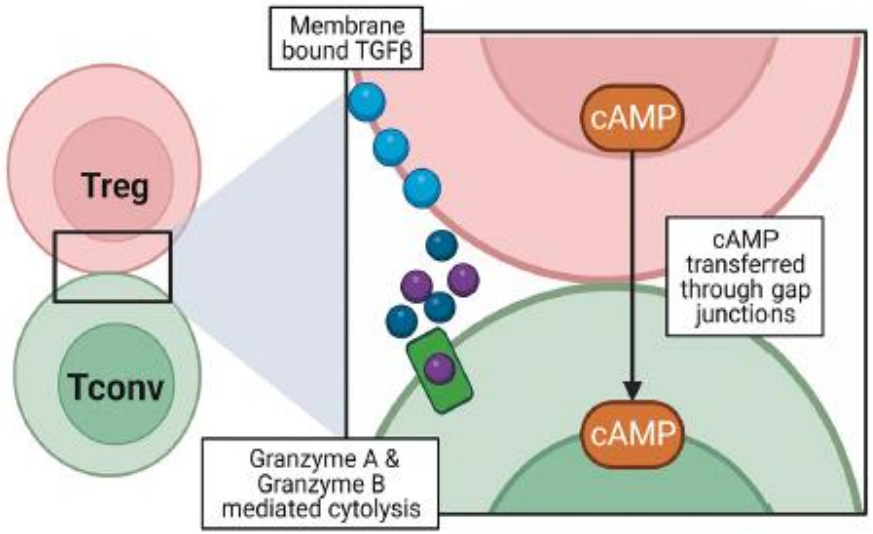


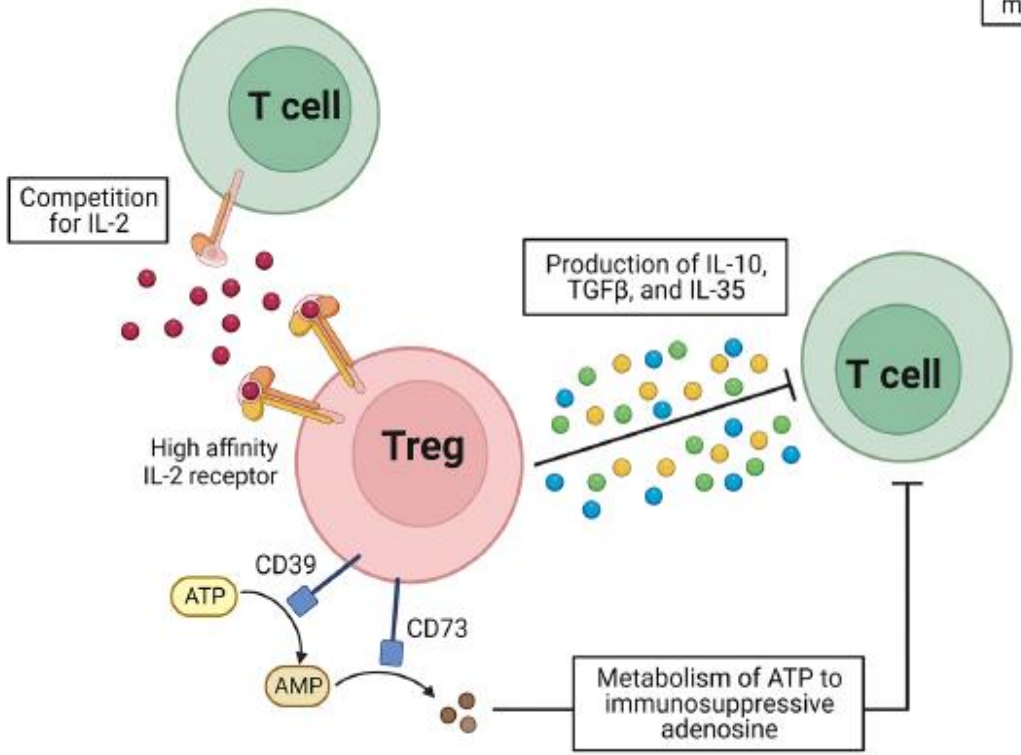
- Specialised Foxp3^{+} Tregs in the B cell follicle
- Absence of Tfr: expansion of Tfh and GC B cells and increase in antibody production
- Tfr promote high affinity antibodies/ limit overall expansion of B cells: fine tuning?
- CXCR5, Bcl-6, CD25+ (or CD25-)

Suppression mechanisms

- Indirect (via effects on other cells, such as DC)
- Direct: cell – cell contact (with conventional T cells)
- Direct (via soluble mediators/ effect on environment): contact independent


Tconv
Production of IDO
CD28
Treg
CTLA-4
IDO
DC
CD80/CD86
Trogocytosis of CD80/CD86
Inhibition of DC activation and function
MHC-II
LAG-3
Treg


Membrane bound TGFβ
Treg
cAMP
cAMP transferred through gap junctions
Tconv
cAMP
Granzyme A & Granzyme B mediated cytolysis

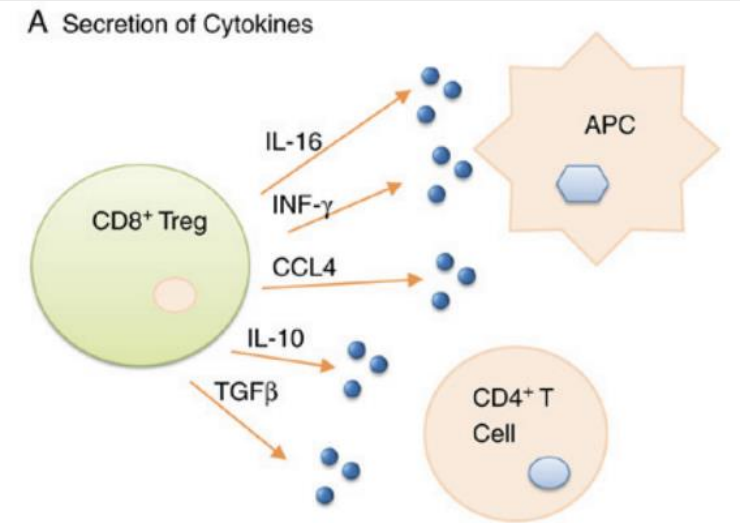

T cell
Competition for IL-2
Production of IL-10, TGFβ, and IL-35
T cell
High affinity IL-2 receptor
Treg
CD39
ATP
CD73
AMP
Metabolism of ATP to immunosuppressive adenosine

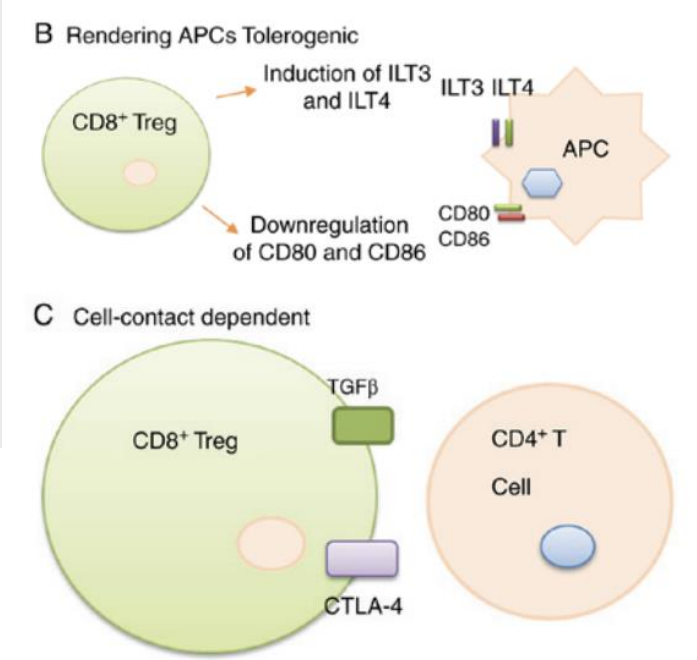
Other Regulatory T cells

- Tr1 (can be generated in vitro by activation in the presence of IL-10; suppress via IL-10 production, or by granzyme dependant killing of APC; express LAG3& CD49β)
- Th3 (CD4+ T cells induced by low-dose tolerance protocols in vivo; produce TGF-β)
- $CD8^+$ suppressor cells (for example $Foxp3^+CD8^+$ or $Ly49^+$/ KIR^+CD8^+)

$CD8^+$ $CD25^+$ $FOXP3^+$ Regulatory T cells

- Low affinity to autologous antigens (not deleted during thymic negative selection):
- CTLA4, CD39
- Less cytotoxic, immunosuppressive cytokines
- Limit immunopathology without affecting viral clearance





Yu et al. *Oncol Lett.* **(2018)**

KIR+CD8+ regulatory T cells

- Similar to mouse Ly49+
- Helios and CXCR3
- Suppress (kill) pathogenic T cells in chronic inflammation
- Autoimmune disease
- Limit the damage induced by autoreactive T cells that are cross-reactive to foreign antigens

J. Li et al., *Science* **376**, eabi9591 (2022)

Are Regulatory T cells clinically relevant?

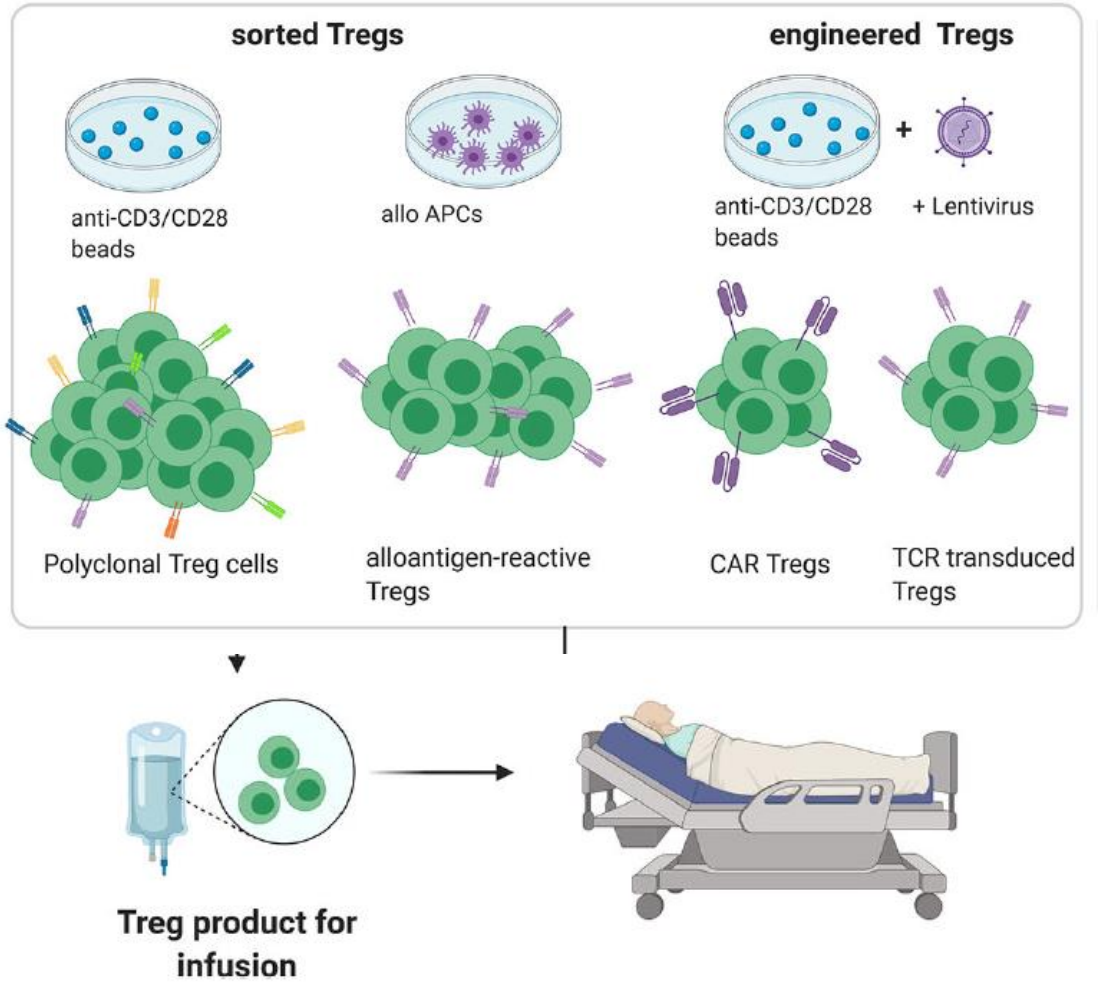
- Wide variety of autoimmune diseases are associated with defects in Treg function (Multiple sclerosis, Rheumatoid arthritis, Type I diabetes, psoriasis)

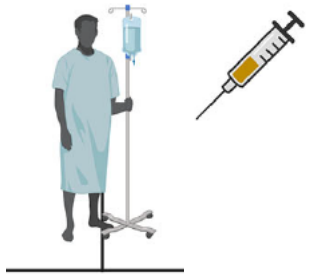
- In transplant patients- positive correlation between graft survival and the number of circulating Tregs

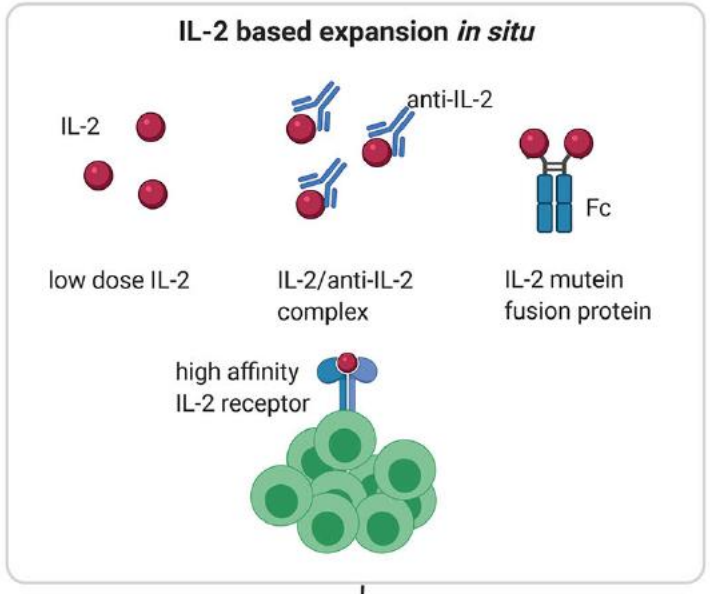
- In cancer patients increased Treg activity associated with poor immune response to tumour antigens

Increase activity / number or decrease activity/ number

Main strategies for Treg therapies







Pilat and Sprent, Front Immunol 2021, Sakaguchi et al Annu Rev Immunol 2021
MacDonald et all Curr opin Biotech 2022

Adoptive Treg therapy hurdles

- Isolate the right cell population (pure!)
  - Magnetic isolation, FACS sorting
  - Which combination of molecules maintains Treg stability and purity
- Collect sufficient number
  - How can you expand them
- Will they be stable once in vivo?
  - ability of cytokines such as IL-1, IL-6 and IL-12 to downregulate Foxp3 expression, suggest that Tregs retain some functional plasticity

Q: Specificity, persistence, function, immunogenicity

Tregs in the clinic: current approaches

- Adoptive Treg therapy
  - Phase I clinical trial: T1D/ Kidney transplant patients
    - Ex vivo Expanded Autologous CD4+CD127lo/-CD25+ Polyclonal Regulatory T Cells
  - Phase I/2 clinical trial: HSCT recipients- to prevent GVHD
    - Freshly isolated cord blood Tregs expanded in vitro
    - Donor Tregs infusion
  - Liver transplant patients
    - Donor alloantigen reactive Tregs
- Adoptive Treg therapy using Tr1 cells (Tr1 cells generated in vitro and re-infused)
    - In HSCT to prevent acute GVHD
    - Crohn's disease
- Adoptive Treg therapy: CAR-Tregs
    - Kidney and liver transplantation

>50 clinical trails (adoptive cell therapy) completed or ongoing

Bluestone JA, McKenzie BS, Beilke J and Ramsdell F (2023) Opportunities for Treg cell therapy for the treatment of human disease. Front. Immunol. 14:1166135.doi: 10.3389/fimmu.2023.1166135

Immune regulation- beyond professional Tregs

- Other cells produce regulatory cytokines
  - IL-10 effects DCs- downregulates co-stim molecules
  - IL-10 can induce anergy in CD4+ T cells
  - IL-10 producing Bregs
  - Regulatory macrophages (Mreg)
- Myeloid-derived suppressor cells (MDSC)
- Anergic cells can affect DC activation
- Inhibitory enzymes: IDO, arginase

Summary

- Responses to autoantigens and to foreign antigens must be regulated in the periphery
- Regulation of immune responses occurs in several forms: apoptosis /activation induced cell death; induction of anergy; production of inhibitory cytokines and Treg mediated suppression
- Regulatory T cells play a crucial role in the control of immune responses and represent great potential target for therapeutic approaches